i. **Dissolved in plasma** About 7 – 10% of CO_2 is transported in a dissolved form in the plasma.

ii. **Bound to haemoglobin** About 20 – 25% of dissolved CO_2 is bound and carried in the RBCs as carbaminohaemoglobin (Hb CO_2)

$$CO_2 + Hb \rightleftharpoons Hb\ CO_2$$

iii. **As bicarbonate ions in plasma** about 70% of CO_2 is transported as bicarbonate ions

This is influenced by pCO_2 and the degree of haemoglobin oxygenation. RBCs contain a high concentration of the enzyme, carbonic anhydrase, whereas small amounts of carbonic anhydrase is present in the plasma.

At the tissues the pCO_2 is high due to catabolism and diffuses into the blood to form HCO_3^- and H^+ ions. When CO_2 diffuses into the RBCs, it combines with water forming carbonic acid (H_2CO_3) catalyzed by carbonic anhydrase. Carbonic acid is unstable and dissociates into hydrogen and bicarbonate ions.

Carbonic anhydrase facilitates the reaction in both directions.

$$CO_2 + H_2O \underset{\text{carbonic anhydrase}}{\rightleftharpoons} H_2CO_3$$

$$\underset{\text{carbonic anhydrase}}{\rightleftharpoons} HCO_3^- + H^+$$

The HCO_3^- moves quickly from the RBCs into the plasma, where it is carried to the lungs. At the alveolar site where pCO_2 is low, the reaction is reversed leading to the formation of CO_2 and water. Thus CO_2 trapped as HCO_3^- at the tissue level it is transported to the alveoli and released out as CO_2. Every 100mL of deoxygenated blood delivers 4mL of CO_2 to the alveoli for elimination.

Bohr effect and Haldane effect

Increase in pCO_2 and decrease in pH decrease the affinity of haemoglobin for oxygen and shifts the oxyhaemoglobin dissociation curve to the right and facilitates unloading of oxygen from hemoglobin in the tissue. This effect of pCO_2 and pH on the oxyhaemoglobin dissociation curve is called the **Bohr effect**.

The **Haldane effect**, on the other hand describes how oxygen concentrations determines hemoglobin's affinity for carbon dioxide. The amount of carbon dioxide transported in blood is remarkably affected by the degree of oxygenation of the blood. The lower the partial pressure of O_2 lower is the affinity of haemoglobin saturation with oxygen hence more CO_2 is carried in the blood. This phenomenon is called **Haldane effect**. This effects CO_2 exchanges in both the tissues and lungs.

In the lungs the process is reversed as the blood moves through the pulmonary capillaries, its pCO_2 declines from 45mm Hg to 40mm Hg. For this to occur carbondioxide is freed from HCO_3^- ions and Cl^- ions moves in to the plasma and reenters the RBC and binds with H^+ to form carbonic acid which dissociates in to CO_2 and water. This CO_2 diffuses along its partial gradient from the blood to the alveoli (Figure 6.8).

6.6 Regulation of Respiration

A specialised respiratory centre present in the medulla oblongata of the hind brain called **respiratory rhythm centre** is responsible for this regulation. **Pneumotaxic centre** present in pons varoli region of the brain moderates the function

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